Alex I.

SOFTWARE ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English – B2+
Polish – B1
Swedish – A2

Domains
Aerospace Systems,
Telecommunications,
Telemetry,
Library Systems,
Oil and Gas Industry.

Software engineer with 13+ years of commercial experience.
C/C++ Software engineer specializing in embedded systems, real-time software development, and hardware-software integration. Experienced in developing, maintaining, and verifying C/C++ software for VxWorks, embedded Linux, and FreeRTOS environments. Strong background in unit and component testing with GoogleTest/GoogleMock, code coverage analysis using GCOV/LCOV, requirements-based verification, static analysis, and code reviews. Recent experience includes developing onboard telecommand and telemetry processing components for an Earth observation satellite, working with ECSS standards, MISRA C++ coding guidelines, CAN communication, and target hardware validation. Additional expertise includes industrial telemetry systems, sensor data acquisition, embedded communication protocols, and multithreaded software development. Experienced in collaborating with cross-functional engineering teams to investigate complex software and hardware integration issues.

Programming Languages
C++, C, C#, Java, JavaScript, Scala, HTML, CSS.

Operating Systems
Windows, Windows Server, Linux, FreeRTOS, VxWorks.

Tools
Jenkins, Coverity, Yocto Project, Qt Designer, Qt Designer .ui forms, systemd, CMake, qmake, gcc, GDB, Valgrind, Qt Test, Visual Studio, Qt Creator, GCOV, LCOV.

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Frameworks & Libraries
GStreamer, Qt, Qt Widgets, QSS, .NET, JUnit, GTest, Node.js, WinForms, GMock, PCL, V4L2, jQuery, Open3D, Qt DBus, Qt WebSockets.

Multimedia
WebRTC, Qt Multimedia, Linphone-SDK, liblinphone.

Databases
MySQL, PostgreSQL, SQLite, Microsoft SQL Server.

Network
SDP, Nginx, SIP, RTP, SNMP, DHCP, TFTP.

Cloud
AWS (CloudWatch, EC2, S3, IAM, RDS, CodeDeploy).

Automotive
CAN.

Embedded
STM32, BitBake, LEON3FT SPARC V8, NXP i.MX8M Mini, IMU, RTK GNSS, RS485.

Development Processes
Jira, Confluence, IBM DOORS.

Source control systems
Git, GitLab, Bitbucket.

Other
JSON, MISRA C++, ECSS-E-ST-40C, ECSS-E-ST-70-41C PUS, EKF, PPS time synchronization.

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Projects

ONBOARD COMMUNICATION SOFTWARE FOR AN EARTH OBSERVATION SATELLITE

Onboard communication flight software for an Earth remote sensing satellite, running on a fault-tolerant LEON3FT SPARC V8 target under VxWorks. The software handles telecommand and telemetry processing and exchanges data with payload controllers over an internal CAN network, subject to project-specific timing requirements. Tested using target hardware benches and software-based test environments to evaluate communication reliability and timing behavior before flight qualification.

Project roles
Embedded Software Engineer

Period
12.2024 – till now

Responsibilities

- Implemented C/C++ telecommand and telemetry processing components using mission-defined PUS packet formats and handling rules;
- Developed and tested C/C++ communication components for message exchange with payload controllers over the onboard CAN network;
- Implemented and tested VxWorks tasks and inter-task communication using message queues, evaluating timing behavior under simulated bus traffic;
- Developed GoogleTest/GoogleMock unit, component, and integration tests for C/C++ communication components, analyzing line and branch coverage using GCOV/LCOV against project verification targets;
- Applied project-specific MISRA C++ coding guidelines and resolved Coverity findings reported through the GitLab CI pipeline;
- Traced verification test cases to functional requirements in IBM DOORS and Jira across the development lifecycle;
- Supported flight-bench validation sessions, helping isolate CAN bus and timing issues against target hardware.

Environment
C++, C, Linux, VxWorks, Coverity, CMake, GCOV, LCOV, GTest, GMock, CAN, LEON3FT SPARC V8, Jira, Confluence, IBM DOORS, Git, GitLab, JSON, MISRA C++, ECSS-E-ST-40C, ECSS-E-ST-70-41C PUS.

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SIP DESK PHONE PLATFORM FOR AUDIO AND VIDEO COMMUNICATION

A SIP desk phone platform for audio and video communication, supporting both camera-enabled and audio-only device models. The project included the native phone application, web management interface, automated provisioning tools, and embedded system customization. It covered everyday calling workflows such as SIP registration, incoming and outgoing calls, video calls, call hold, transfer, conferencing, phonebook access, direct IP dialing, remote configuration, and device setup.

Project roles
Software Engineer

Period
06.2022 – 12.2024

Responsibilities

- Refactored SIP call handling toward a cleaner MVC/MVVM-style architecture by separating UI rendering from call state, user actions, media negotiation, and Linphone-SDK integration logic;
- Incrementally decomposed legacy Qt Widgets call-flow code into clearer domain, presentation, and SDK-integration boundaries while preserving existing UI behavior and QObject signal compatibility;
- Hardened video-call and conference UI behavior for edge cases such as held calls, second-line calls, remote hangups, stale overlays, and active video stream switching;
- Contributed to Linphone-SDK (liblinphone) backend customization for SIP failover and backup-server connectivity, improving device reliability;
- Developed WebRTC video-conferencing UI flows: conference creation and join, incoming invite screens, participant roster, status strip, control panel, video tile layout, and audio/video device controls;
- Unified project file formats across the device ecosystem, and implemented provisioning workflows using SNMP, DHCP, TFTP, and SQLite;
- Collaborated on Yocto/BitBake customization, i.MX embedded Linux target integration, and device-side service configuration for hardware-software integration.

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Environment
C++, C#, JavaScript, HTML, CSS, Windows, Linux, Yocto, Qt Designer, Qt Designer .ui forms, systemd, qmake, gcc, gdb, Valgrind, QtTest, Qt Creator, GStreamer, Qt, Qt Widgets, QSS, .NET, Node.js, WinForms, V4L2, Qt DBus, Qt WebSockets, WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, SQLite, SDP, SIP, RTP, SNMP, DHCP, TFTP, BitBake, NXP i.MX8M Mini, Jira, Git, GitLab.

UAV-BASED 3D RECONSTRUCTION SOFTWARE FOR LARGE-SCALE OBJECTS

A software platform for automated 3D reconstruction of large objects such as buildings and industrial facilities using UAV-mounted optical and laser sensors. The system processes aerial imagery and 2D laser scans captured along the flight path and turns them into dense point clouds and accurate 3D models. A dedicated Software-in-the-Loop environment reproduces flight paths and sensor behavior on the bench, so reconstruction logic can be validated before any aircraft leaves the ground. It gives surveying, construction and quality control teams reliable spatial data on the whole structure without manual on-site measurement

Project roles
Software Engineer

Period
03.2019 – 06.2022

Responsibilities
- Developed software modules for digital image processing systems used in Unmanned Aerial Vehicles, forming the data acquisition layer for aerial 3D reconstruction of large structures;
- Aligned laser, camera, and navigation data using GNSS PPS-based timing and sensor timestamps;
- Integrated EKF-derived RTK GNSS/IMU trajectory estimates and sensor streams into the data ingestion pipeline;
- Processed and inspected the resulting point clouds with PCL and Open3D, filtering and downsampling large-area scans and visualizing them for review;
- Created conversion tools between the payload's capture formats and the formats used downstream in the reconstruction pipeline;
- Built a Software-in-the-Loop environment that ran payload software against simulated UAV flight paths, letting the team validate reconstruction pipelines without booking real flights;
- Emulated point-cloud sensors inside the SIL rig, generating synthetic laser returns with realistic noise and drop-outs to exercise the reconstruction stack against edge cases;

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- Paired with the flight control and surveying teams to agree sensor mounting, capture patterns and acceptance criteria for each scan campaign.

Environment
C++, Linux, CMake, PCL, Open3D, IMU, RTK GNSS, Jira, Confluence, Git, Bitbucket, EKF, PPS time synchronization.

LIBRARY MANAGEMENT SYSTEM

A robust and highly scalable library management system with over 40 years of operational history, serving a diverse client base across the globe. The system is built using a combination of C, Java, and Scala, leveraging the strengths of each language to deliver a reliable, efficient, and feature-rich platform. Designed to handle high volumes of data and complex workflows, it has continually evolved to meet the needs of modern libraries while maintaining compatibility with legacy components.

Project roles
Software Engineer

Period
09.2016 – 03.2019

Responsibilities
- Served as the main C contributor on the core transaction engine, delivering circulation and cataloguing features the client's largest sites depended on;
- Diagnosed and resolved long-standing defects in the legacy C codebase, tracing memory and concurrency issues under gdb against production-like datasets;
- Refactored existing Scala API code to improve maintainability and scalability;
- Automated build and deployment by wiring Jenkins builds into AWS CodeDeploy releases onto EC2, with artifacts versioned in S3, removing manual steps from each release and shortening development cycles;
- Modified database tables to support new feature implementations, improving schema design to align with updated system workflows;
- Conducted thorough code reviews across both the in-house and the client's teams, holding the line on standards and spreading context on legacy subsystems;
- Estimated C-side tasks, keeping milestones realistic for a system carrying four decades of accumulated behavior.

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Environment
C, Java, Scala, Linux, Jenkins, gdb, JUnit, MySQL, PostgreSQL, Amazon CloudWatch, Amazon EC2, Amazon S3, Amazon RDS, Jira, Confluence, Git, Bitbucket, AWS (IAM, CodeDeploy).

CORPORATE SYSTEM FOR THE OIL AND GAS INDUSTRY

A corporate telemetry system for oil and gas field operations, covering both the equipment side and the office side. Field units collect readings from pressure, flow and temperature sensors over industrial buses with periodic sampling and timing requirements, protect critical measurements on the device itself, and forward them to a central server. Engineers and managers work with the same data through an internal web application, which makes well performance and equipment health visible across sites without waiting for manual reports.

Project roles
Software Engineer

Period
06.2013 – 09.2016

Responsibilities

- Developed a real-time C++ data acquisition module for pressure, flow, and temperature sensors, using periodic sampling and timing-aware processing;
- Interfaced directly with field hardware over CAN and RS-485, bringing up sensor boards on the bench and tracing signal-level faults together with the instrumentation engineers;
- Implemented on-device protection of critical measurement data, combining checksums with redundant storage so telemetry survived power loss and link outages at remote sites;
- Developed embedded C++ components following project-specific reliability and error-handling requirements, taking documented failure scenarios into account;
- Engineered the back-end services that ingested and normalized field telemetry, keeping ingestion stable as the number of monitored wells grew;
- Built the internal web front end with JavaScript, HTML and jQuery, giving field and office staff a shared view of live readings and equipment status;
- Maintained and enhanced the existing codebase by resolving bugs, optimizing performance, and reducing technical debt;
- Removed outdated legacy components, modernizing the system to improve reliability and maintainability;

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- Coordinated with the hardware team on firmware and protocol changes, keeping the device and server sides in step across releases.

Environment

C++, JavaScript, HTML, Windows Server, FreeRTOS, CMake, Visual Studio, GTest, jQuery, MySQL, Microsoft SQL Server, Nginx, CAN, STM32, RS485, Git, Bitbucket.

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	C++	11	2026
	C	4	2026
	C#	3	2024
	Java	2	2019
	JavaScript	6	2024
	Scala	2	2019
	HTML	6	2024
	CSS	3	2024
Operating Systems	Windows	3	2024
	Windows Server	3	2016
	Linux	10	2026
	FreeRTOS	3	2016
	VxWorks	2	2026
Tools	Jenkins	2	2019
	Coverity	2	2026
	Yocto Project	3	2024
	Qt Designer	3	2024
	Qt Designer .ui forms	3	2024
	systemd	3	2024
	CMake	8	2026
	qmake	3	2024
	gcc	3	2024
	GDB	5	2024
	Valgrind	3	2024
	Qt Test	3	2024
	Visual Studio	3	2016

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	Qt Creator	3	2024
	GCOV	2	2026
	LCOV	2	2026
Frameworks & Libraries	GStreamer	3	2024
	Qt	3	2024
	Qt Widgets	3	2024
	QSS	3	2024
	.NET	3	2024
	JUnit	2	2019
	GTest	5	2026
	Node.js	3	2024
	WinForms	3	2024
	GMock	2	2026
	PCL	3	2022
	V4L2	3	2024
	jQuery	3	2016
	Open3D	3	2022
	Qt DBus	3	2024
	Qt WebSockets	3	2024
Multimedia	WebRTC	3	2024
	Qt Multimedia	3	2024
	Linphone-SDK	3	2024
	liblinphone	3	2024
Databases	MySQL	6	2019
	PostgreSQL	2	2019
	SQLite	3	2024
	Microsoft SQL Server	3	2016

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Network	SDP	3	2024
	Nginx	3	2016
	SIP	3	2024
	RTP	3	2024
	SNMP	3	2024
	DHCP	3	2024
	TFTP	3	2024
Cloud	AWS	2	2019
Automotive	CAN	5	2026
Embedded	STM32	3	2016
	BitBake	3	2024
	LEON3FT SPARC V8	2	2026
	NXP i.MX8M Mini	3	2024
	IMU	3	2022
	RTK GNSS	3	2022
	RS485	3	2016
Development Processes	Jira	10	2026
	Confluence	8	2026
	IBM DOORS	2	2026
Source control systems	Git	13	2026
	GitLab	4	2026
	Bitbucket	9	2022
Other	JSON	2	2026
	MISRA C++	2	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	ECSS-E-ST-40C	2	2026
	ECSS-E-ST-70-41C PUS	2	2026
	EKF	3	2022
	PPS time synchronization	3	2022

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